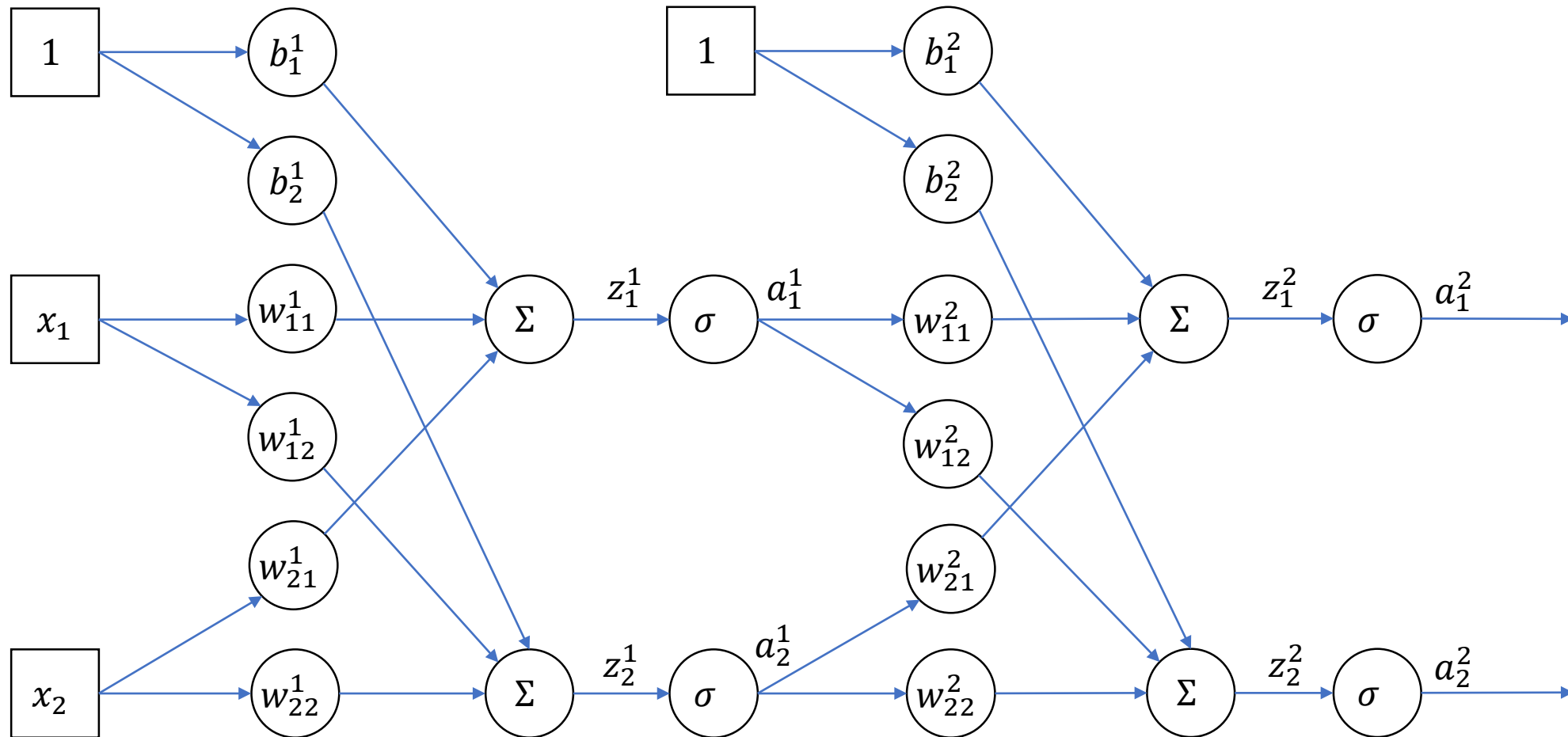


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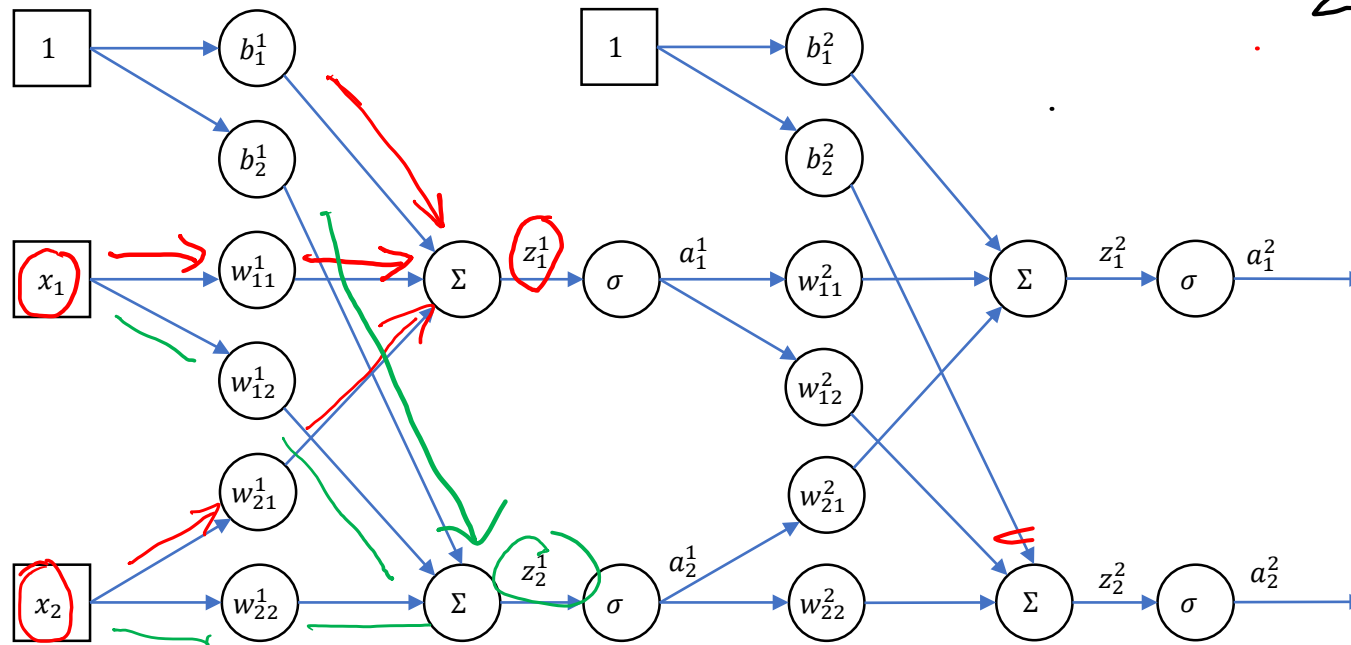
The forward pass



Breakdown



Forward pass



$$z_1^1 = w_{11}^1 x_1 + w_{21}^1 x_2 + b_1^1$$

$$z_2^1 = w_{12}^1 x_1 + w_{22}^1 x_2 + b_2^1$$

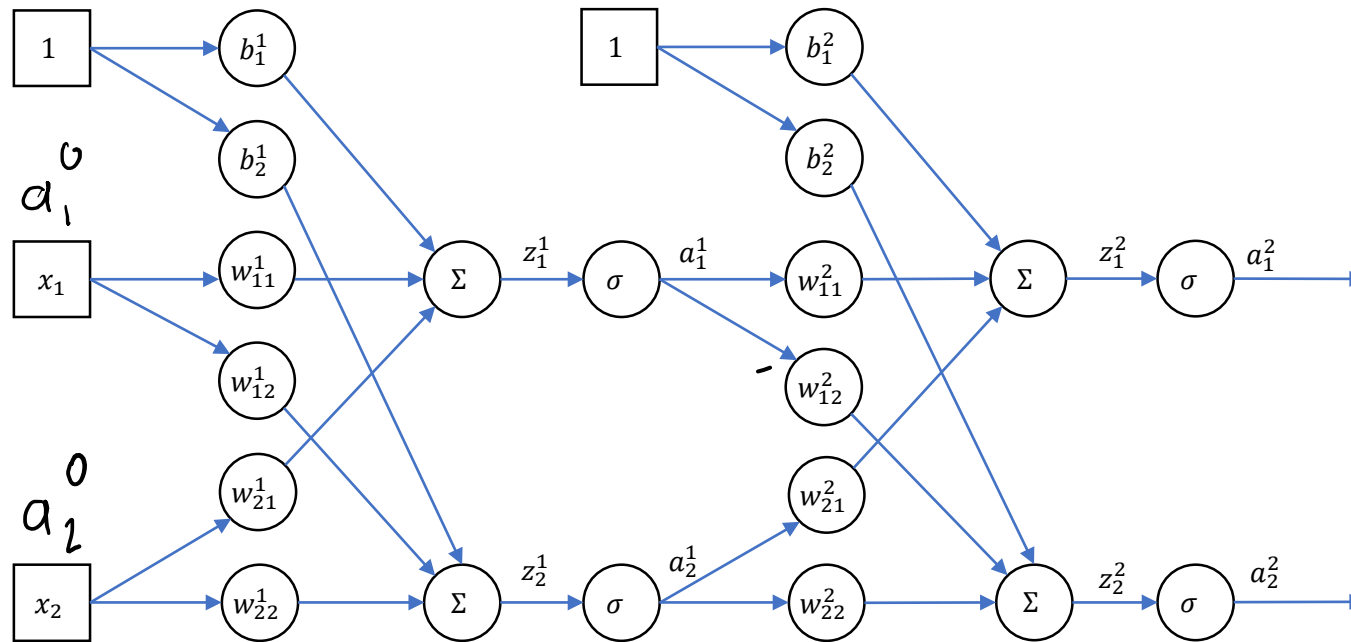
$$W = \begin{bmatrix} w_{11} & w_{12} \\ w_{21} & w_{22} \end{bmatrix}$$

$$W^T = \begin{bmatrix} w_{11} & w_{21} \\ w_{12} & w_{22} \end{bmatrix}$$

$$\mathbf{z}^1 = \begin{bmatrix} z_1^1 \\ z_2^1 \end{bmatrix} = \begin{bmatrix} w_{11}^1 & w_{21}^1 \\ w_{12}^1 & w_{22}^1 \end{bmatrix} \begin{bmatrix} x_1 \\ x_2 \end{bmatrix} + \begin{bmatrix} b_1^1 \\ b_2^1 \end{bmatrix}$$



Forward pass



$$\mathbf{z}^1 = (W^1)^T \cdot \mathbf{x} + \mathbf{b}^1$$

$$\mathbf{a}^1 = \sigma(\mathbf{z}^1) = \begin{bmatrix} \sigma(z_1^1) \\ \sigma(z_2^1) \end{bmatrix}$$

$$\mathbf{z}^2 = (W^2)^T \cdot \mathbf{a}^1 + \mathbf{b}^2$$

$$\mathbf{a}^2 = \sigma(\mathbf{z}^2) = \begin{bmatrix} \sigma(z_1^2) \\ \sigma(z_2^2) \end{bmatrix}$$

$$\mathbf{z}^l = (W^l)^T \cdot \mathbf{a}^{l-1} + \mathbf{b}^l$$



Next up:

Forward pass numerical example

